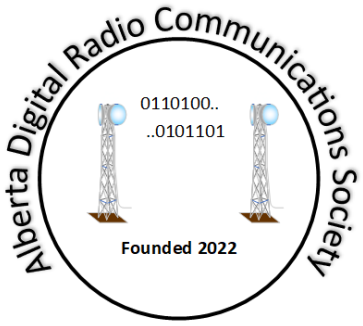


Digital Radio Projects

Interfacing Radios to AMNC

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[1] gnu.org, "General Public Licence," [Online]. Available: https://www.gnu.org/licenses/gpl-3.0.en.html. [Accessed 25th February 2018].

Revision Status

Revision	Date	Description
0.1	Sept 28, 2026	Initial Draft
0.2	Oct 20th, 2026	Minor updates

Table 1 Revision status

Reference Documents

Author	Document	Description
M. Alcock	IP400-UFP-OFDM	IP400 UFP for OFDM based nodes

Table 2 Reference Documents

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AMNC Connector

The AMNC DB-15 connector has three distinct sets of connections, as listed in Table 3 and illustrated in Figure 1

Pin	Connection	Pin	Connection
1	Port 1 Audio Out	**8**	Port 2 PTT
2	Port 2 Audio Out	**10**	Port 2 Carrier Detect
3	Port 1 Carrier Detect	**11**	Port 1 Audio In
4,9,14	Power Ground	**12**	Port 2 Audio In
5, 15	+12V	**13**	Port 1 PTT
6	Radio Ground	**7**	Radio Ground

Table 3 AMNC DB-15 Pinout

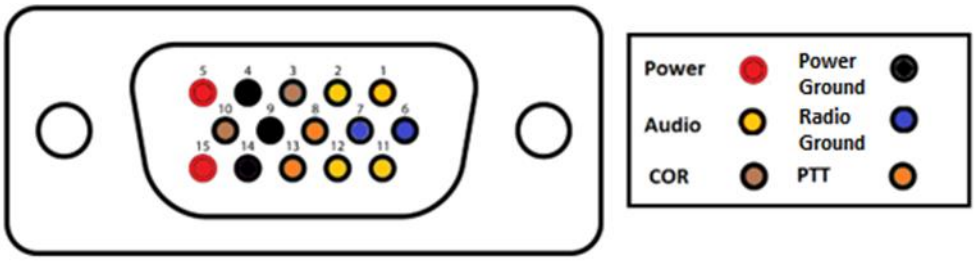


Figure 1 AMNC Connector

Power Connection

The power connections are used only when the AMNC is mounted on a raspberry Pi, to power itself as well as providing a +5V supply to the Pi. In the standalone mode, +5v power is supplied by the USB connector, and the power connections on the DB-15 connector are not used.

Radio Ports

Each radio port has an audio input and output, a radio ground, a carrier detect input and PTT output. The carrier detect can be positive or negative, which is configurable in the radio setup menu. An LED shows the status of the input voltage from the radio, not the carrier status. The PTT is an open drain FET that is ground activated and can sink up to 60 mA.

Radio Breakout Cable

Two standards have been adopted for the radio breakout, the first is a 'Y' cable that connects the AMNC's DB-15 connector to two individual DB-9 connectors, one for each radio port. The pinout of the DB-9 connectors is the same. The DB-9 connector part number from DigiKey is 4672-MHDB9SP-ND, and the DB-15 connector is AE11037-ND.

DB9 Standard

Table 4 and Table 5 illustrate the wiring of the radio breakout cable.

Port 1

AMNC DB-15 Pin	DB-9 Pin	Wire Colour	Purpose
5	1	White	+13VDC Input[1]
13	2	Green	PTT
4	5	Black	Power Ground
11	6	Blue	Audio in to modem
3	7	Red	Carrier Detect
1	9	Yellow	Audio out to radio

Table 4 DB9 to AMNC mapping for port 1

Port 2

AMNC DB-15 Pin	DB-9 Pin	Wire Colour	Purpose
15	1	White	+13VDC Input
8	2	Green	PTT
14	5	Black	Power Ground
12	6	Blue	Audio in to modem
10	7	Red	Carrier Detect
2	9	Yellow	Audio out to radio

Table 5 DB9 to AMNC mapping for port 2

Modular Jack Standard

A standard for easy connection to a radio is a 6-conductor cable with an RJ-11 modular jack. The radio can power the AMNC so that both can power up together. Table 6 illustrates the connections.

Pin[2]	Wire Colour	Usage	Purpose
1	White	VCC	Power input, +13VDC
2	Black	GND	Ground
3	Red	CARR	0V when carrier present, 5V otherwise
4	Green	PTT	Low active push-to-talk output
5	Yellow	Modulator	Connects to the radio modulator
6	Blue	Discriminator	Connects to the radio discriminator

[1] Power is sourced externally only when the AMNC is used in the Pi HAT mode
[2] Tab down, pin 1 is on the left hand side

Table 6 Modular Jack connections

Interconnect Module

The radio interface module provides a connection between the DB-9 radio breakout cable and the modular jack cable, with a circuit to add up to ±26dB of gain in either path. The module is powered by the +13V connection, which is sourced from the radio.

Figure 2 Illustrates the radio interconnect module.

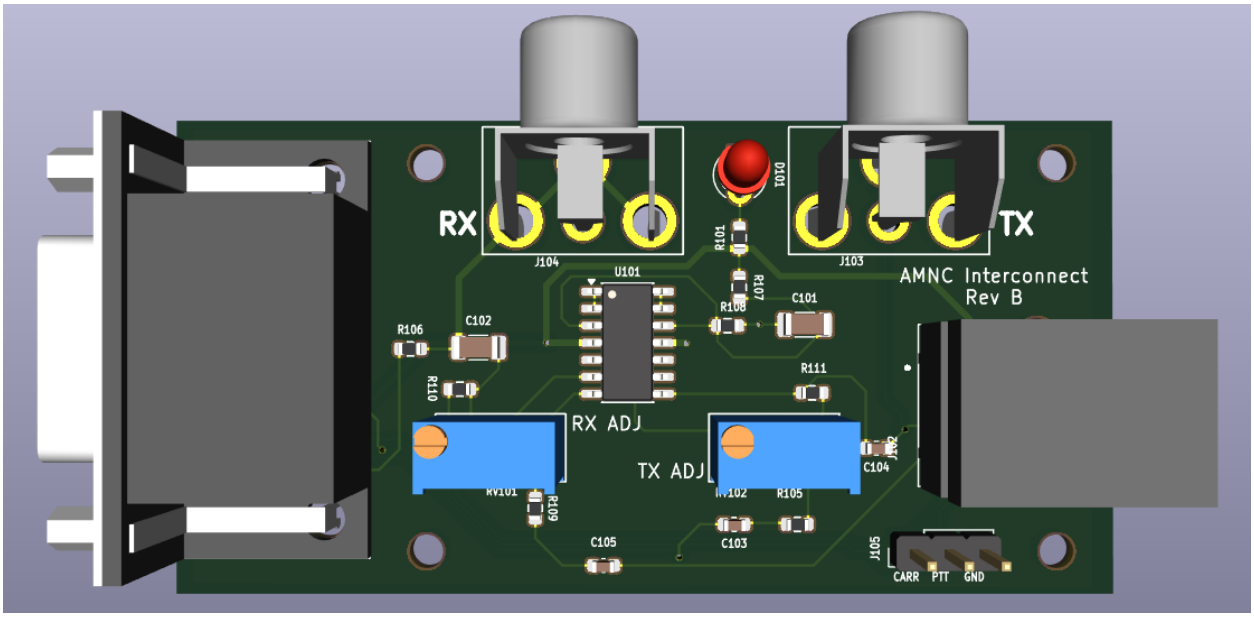


Figure 2 Interface Module

Connection

The module conforms to the DB-9 standard facing the AMNC, and the 6-pin modular jack standard facing the radio. The VCC power for the op amp can be taken from the radio or AMNC side, the maximum supply voltage is 16v, and it operates from rail to rail.

Table 7 lists the module specifications.

Parameter	Conditions	Min	Nom	Max
VCC	Either port	6	12	16
Gain	±5%	-24.6dB	0	24.6dB
Input Voltage	Max gain, either path, max VCC			942mV
	Max gain, either path, nom VCC			700mV
Output Voltage	Input voltage 1.00V, any VCC	59mV	1.0V	VCC

Table 7 Interconnect Module Specifications

Pinout

The modular jack connection is the same as that in Table 6, the DB-9 connector pinout is illustrated in Table 8. The recommended cable can be purchased from DigiKey, part number A3662R-07-ND, or from Assmann WSW components, part number AT-S-26-6/6/B-7-OE.

Pin	Connection	Pin	Connection
1	+12V	6	Receive Audio from radio
2	PTT (ground active)	7	Carrier detect from radio
3	NC	8	NC
4	NC	9	Transmit audio to radio
5	GND		

Table 8 Interface module DB-9 Connector

In addition, there are several other connection points, as illustrated in Table 9.

Connection	Type	Purpose
Rx Audio	RCA Connector	Monitors the Rx audio after amplification
Tx Audio	RCA Connector	Monitors the Tx audio after amplification
CARR	Pin J105-1	Current state of the carrier detect
PTT	Pin J105-2	Current state of the PTT signal
Gnd	Pin J105-3	Ground

Table 9 Interface board connections

Tune Up

Connect the module to the AMNC port using the radio breakout cable as described in Table 4 for port 1 or Table 5 for port 2. The module requires +12V for proper operation, which is connected to either the DB9 or modular jack connector.

For transmit, enable the 1500 Hz test tone on the AMNC, and adjust the Tx ADJ potentiometer for about 2.5KHz deviation. The actual voltage level will vary between radios.

For receive, inject a 1KHz tone at 2.5KHz deviation, or connect to another radio that has 1500Hz test tone enabled, then adjust the level at the Rx monitor connection for about 1v p-p with the radio unsquelched for about 1V p-p, which can be measured on an oscilloscope, or 0.707Vrms on a voltmeter, or 0dBV on a RMS voltmeter.